

r/aiwars

It takes at least 5 liters of water to make one piece of paper. Drawing on paper uses more water than using an AI generator.

Posted by u/CBrinson • • 0 points • 216 comments

Links below. They all put the estimate over 5 liters. Some significantly so.

The reality is that everything uses water. I mostly just want to show that AI's water usage isn't something to be scared of. It's just a strawman argument. Don't stop drawing because you are scared of using a resource that will literally fall as rain almost as soon as you use it.

Use AI or don't use AI. It's none of my business and I don't judge you either way. I hope you get what you want from whichever you choose and don't feel the need to tell others their choice is invalid -- in either direction.

<https://watercalculator.org/footprint/the-hidden-water-in-everyday-products/>

<https://www.theworldcounts.com/challenges/consumption/other-products/environmental-impact-of-paper>

<https://www.theatlantic.com/technology/archive/2012/06/it-takes-more-than-3-gallons-of-water-to-make-a-single-sheet-of-paper/258838/>

Comments

u/PLMMJ • 1 points • 2026-02-12 22:43 UTC

Except I did the math and AI likely uses up billions more gallons of water than paper.

Part 1: The Mouth-Breather Math

I went to the site Tom's Hardware (because an AI bro cited it), and after a bit of meandering, found [an article](<https://www.tomshardware.com/tech-industry/explosive-ai-buildout-brings-into-question-water-supply-concerns-exploring-how-data-centers-could-curb-water-demands>) on data center water use.

Here it says that each data center can consume over 5 million gallons of water daily. Then, I looked up how many data centers are in the USA. I usually got numbers around 5400. Doing the mouth-breather math from this information only, $5,000,000 \times 5400 \times 365 = 9,855,000,000,000$ (9.855 trillion) gallons of water yearly! That's a lot! But that's for all data centers. What about only AI data centers?

Part 2: Approximating AI data centers

I couldn't find stats on AI data centers, but I could on servers running AI.

Looking at [the report](<https://escholarship.org/uc/item/32d6m0d1>) linked in that article (specifically page 30), they say that of in 2020, of the 20 million installed servers, 1.6 million (8%) of them were AI. Their projection for 2028 is that there will be 37 million installed servers and 8-12 million of them will be AI ($\sim 21.6\%$ - $\sim 32.4\%$) and there's also a graph with a projection for 2024 with ~ 31 million installed servers and ~ 5 -6 million of them being AI ($\sim 16\%$ - $\sim 19\%$). Taking the average of the lower bound of those figures since we're in 2026, we get about 18.9% of servers are AI. Rounding that down to 18% and assuming it's the same as AI data centers compared to non-AI data centers, we get 972 AI data centers in the US. Replacing our earlier 5400 value of total data centers with 972 approximate AI data centers, we still get 1.7739 trillion gallons of water yearly.

Part 3: Comparison to paper

Going off of [more sources](<http://www.mntap.umn.edu/industries/facility/paper/water/>), sources admittedly old for my liking but the best I could get, the US benchmark for paper production is 17,000 gallons per ton of paper. Then, for paper production, I got a global number of 82 million from [this site](<https://www.statista.com/statistics/1089445/global-writing-and-printing-paper-demand/>). Multiply the numbers and we get 1.394 trillion gallons per year, which is a bit over 3/4 of the AI water use number. If we assume that paper worldwide is around the same as the US benchmark, the paper industry uses less water than US AI data centers. (Other countries have a lot less data centers than the US, with Germany as the runner-up at 529 and them plus the rest of the top 10 only having ~ 3200 combined.)

And the thing about the AI number is that according to the original article, the report only covers cooling, ignoring the water used to generate power for the data centers or on-site needs outside of cooling. And it's even more in hotter parts of the US, meaning that this number has a multiplier that would be based on the temperature of every AI data center, which I don't think I can calculate.

u/CBrinson • 1 points • 2026-02-12 23:26 UTC

You literally took the "as much as" number and used it as an average and you think your argument has any validity?

Also your core premise is total usage which is an odd and irrelevant metric. What matters is the amount of paper vs the amount of AI prompts that are equivalent.

You basically just made up your own numbers. Good job I guess?

u/PLMMJ • 1 points • 2026-02-12 23:40 UTC

It's not an "as much as" number, it says "upwards of 5 million" in the original. If the average is 4 million it's still higher than paper.

u/CBrinson • 1 points • 2026-02-12 23:51 UTC

I am kicking you out of this convo because you don't seem to when the intelligence to process basic info.

Because you have to measure the replacement cost of one vs the other if you are comparing the totals.

u/Kilroy898 • 1 points • 2026-01-03 23:57 UTC

This is interesting. I would like to note that while it does use a lot of water to make a single piece of paper, that water is reused extensively. The same gallon will be used 20+ times before new water is added, and most paper mills also recycle their water to get even further use out of it. They don't use a ***New*** gallon for every sheet of paper.

u/raggarganget • 1 points • 2025-12-04 12:24 UTC

The water used for paper production isn't rendered useless though? It goes through the slow cycle of regeneration that makes our 3% freshwater drinkable, regenerative and usable. The literal active poison in the water used for AI is first of all, insanely difficult to remove (fully removing it is almost impossible), and the amount of water used daily is screwing up the natural cycle of water regeneration because of its insane speed, which causes the freshwater to be used up, poisoned, and not able to recuperate.

u/CBrinson • 1 points • 2025-12-04 17:00 UTC

The water from AI is also not rendered useless. Evaporation/distillation removes all contaminants and happens literally to all water just as you describe for water for paper.